Summary

10.21

Added a module called "tap (transport as plugin)" in Signal, changing the transmission of Signal ciphertext from Signal Server to a user-configured provider. Used in the following cases:

1. Private chat:

A initiates "use tap mode", generates a read-only token for A's configured cloud_A, and sends it to B.
After B accepts the request, B also sends a read-only token (of cloud_B) to A.
Then, tap mode is activated:
A sends messages by uploading ciphertext to cloud_A.
B periodically pulls ciphertext from cloud_A and decrypts it.
Similarly, B can send messages via Cloud B, and A pulls from Cloud B using token_B

2. Group chat:

A initiates "use tap mode", generates a read-only token for A's configured cloud_A, and sends it to all members. The group enters “Proposing” state.
After receiving the request, B agrees and sends a read-only token for cloud_B to all members.
...
Once all members' tokens are locally available, the group enters tap mode.
A sends messages by uploading ciphertext to cloud_A.
A periodically pulls ciphertext from other members' clouds, then decrypts them.

The data we upload to the cloud

```
data class TransportMessage(
    val version: String = CURRENT_VERSION,
    val messageId: String,
    val timestamp: Long,
    val senderId: String,
    val recipientId: String,
    val messageType: TransportMessageType,
```

```
    val signalCiphertext: String,  // Base64
    val signalCiphertextType: Int,
    val contentMetadata: TransportContentMetadata,
    val attachments: List<TransportAttachment> = emptyList()
)
```

In the signalCiphertext:

A. Private Chat:

Version Byte – 1 byte, current version is `0x03`
SignalMessage (Protobuf format) contains:

1. ratchetKey – 32 bytes, Diffie-Hellman temporary public key, used in Double Ratchet protocol
2. counter – varint, message chain counter, prevents replay attacks
3. previousCounter – varint, counter for the previous chain
4. ciphertext – variable length, actual encrypted message (AES-CBC), includes padding
5. MAC – 8 bytes, first 8 bytes of HMAC-SHA256, verifies message integrity and authenticity

B. Group Chat:
Version Byte – 1 byte, current version is `0x03`
SenderKeyMessage (Protobuf format) contains:

1. distributionId – 16-byte UUID, unique ID for the group distribution key
2. chainId – uint32, sending chain ID
3. iteration – uint32, iteration counter in the chain, prevents replay attacks
4. ciphertext – variable length, encrypted message content (AES-CBC) with padding
5. signature – 64 bytes, Ed25519 signature to verify sender identity and message integrity

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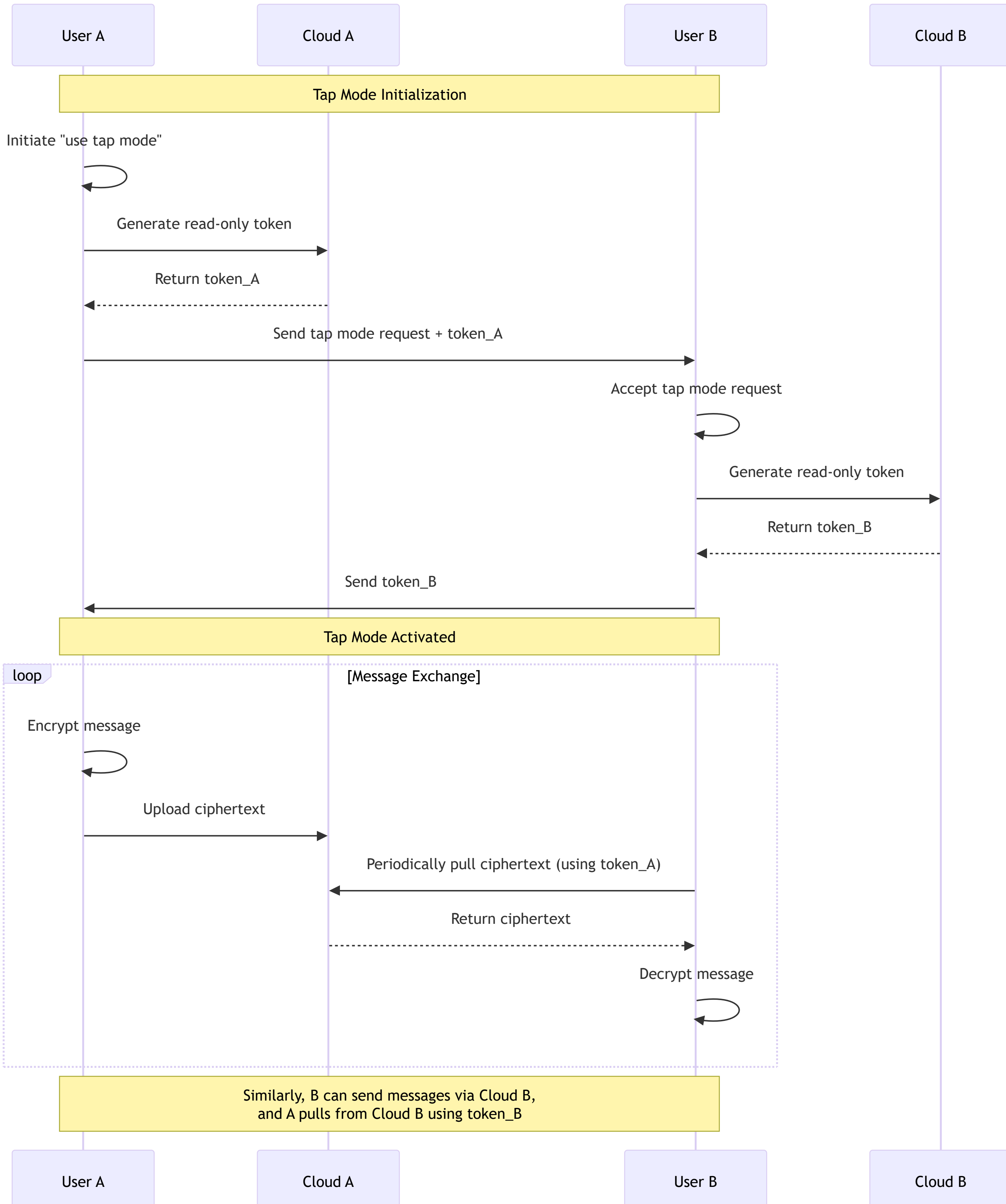
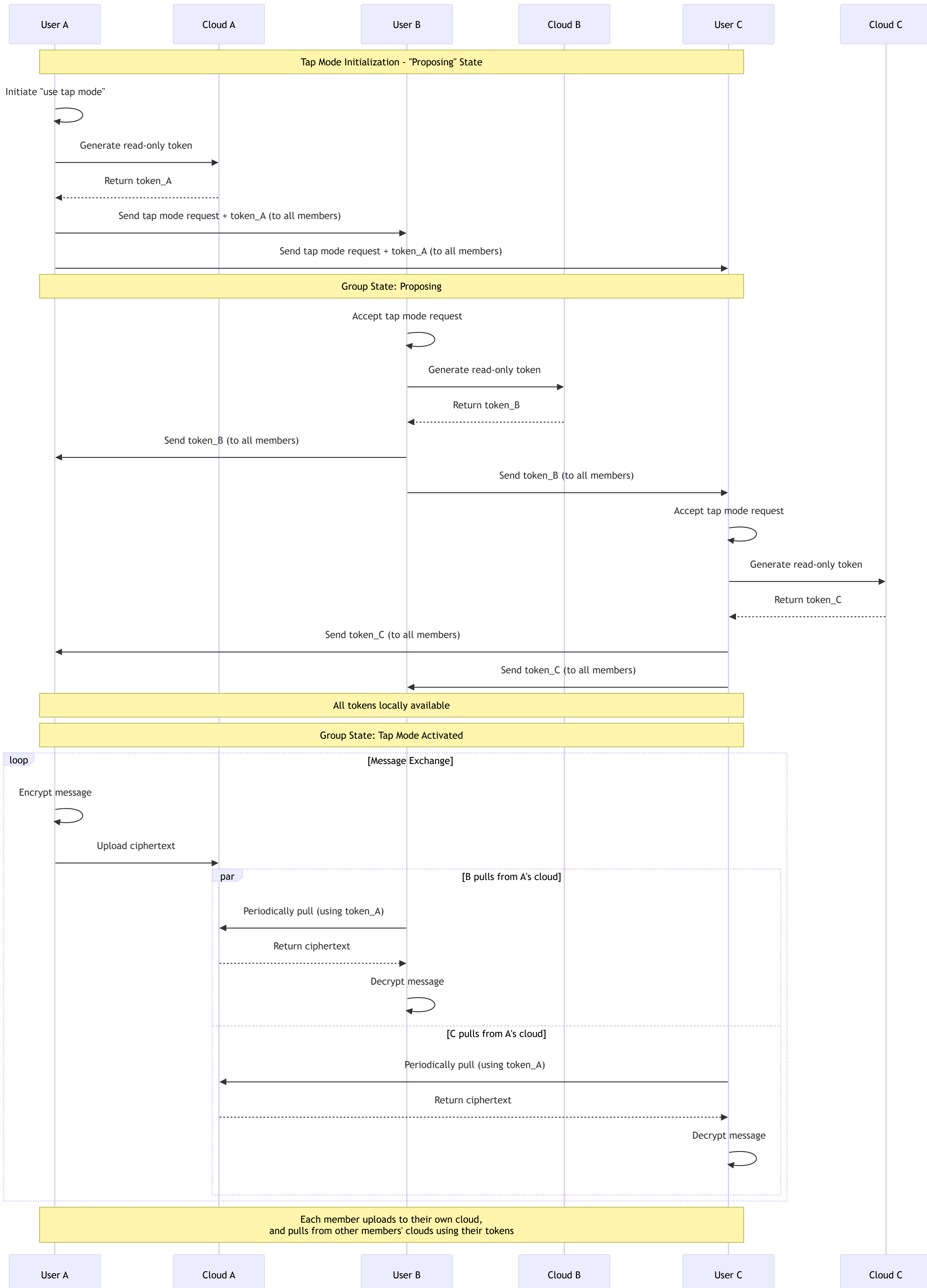
Diagrams

1. Private chat:


User A
Cloud A
User B
Cloud B
Tap Mode Initialization
Initiate "use tap mode"
Generate read-only token
Return token_A
Send tap mode request + token_A
Accept tap mode request
Generate read-only token
Return token_B
Send token_B
Tap Mode Activated
loop
[Message Exchange]
Encrypt message
Upload ciphertext
Periodically pull ciphertext (using token_A)
Return ciphertext
Decrypt message
Similarly, B can send messages via Cloud B,
and A pulls from Cloud B using token_B
User A
Cloud A
User B
Cloud B

2. Group chat:


User A
Cloud A
User B
Cloud B
User C
Cloud C
Tap Mode Initialization - "Proposing" State
Initiate "use tap mode"
Generate read-only token
Return token_A
Send tap mode request + token_A (to all members)
Send tap mode request + token_A (to all members)
Group State: Proposing
Accept tap mode request
Generate read-only token
Return token_B
Send token_B (to all members)
Send token_B (to all members)
Accept tap mode request
Generate read-only token
Return token_C
Send token_C (to all members)
Send token_C (to all members)
All tokens locally available
Group State: Tap Mode Activated
loop
[Message Exchange]
Encrypt message
Upload ciphertext
par
[B pulls from A's cloud]
Periodically pull (using token_A)
Return ciphertext
Decrypt message
[C pulls from A's cloud]
Periodically pull (using token_A)
Return ciphertext
Decrypt message
Each member uploads to their own cloud,
and pulls from other members' clouds using their tokens
User A
Cloud A
User B
Cloud B
User C
Cloud C

Some notes

1. An abstract class has been added; the program no longer directly depends on Tencent or AWS services. Additional storage providers can be added in the future.
2. Only the transport layer was changed; the encryption feature remains unchanged.
3. One finding: Signal’s two-person chat still uses Double Ratchet encryption, not Sender Key as used in other group chats.